

Considering Automotive SBOM

2025.11.14

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What I want to share with you today

To Accelerate Smart and Effective SBOM Usage in Automotive Industry,
we launched a group to establish **Automotive SBOM** as a best practice.

The objectives of this group are to:

- Identify key challenges, concerns, and barriers to effective SBOM usage
- Promote awareness and position Automotive SBOM as best practices across the automotive industry

We want to hear real-world input from the entire supply chain.

Share your insights—join our group today!

Background

Challenges of SBOM Implementation in the Automotive Industry



Automotive SBOM

We have reconsidered the content to focus on operations in the automotive industry, and defined an Automotive SBOM as an SBOM specification that follows the general-purpose SBOM specification.

Purpose of Automotive SBOM

1 Use as a Common Standard in the Supply Chain

- Automotive SBOM defines the format, data content, and granularity of the information that OEMs requires.
- Being used in suppliers as **a common guideline for creating SBOMs that meet quality criteria**, Automotive SBOM improve transparency and traceability throughout our supply chain.

2 Contribution to Improving Productivity in the Automotive Industry

- Automotive SBOM was designed with the intention to be commonly used by OEMs both in Japan and overseas.
- By **standardizing the requirements from OEMs to suppliers**, the burden on suppliers to comply with multiple requests from OEMs will be reduced.

3 Use as a Requirement Specification for Tool Vendors

- The content of SBOM can be collected by using SCA tool, but the functions and performance of the SCA tools currently available on the market are not perfect.
- Automotive SBOM can be used as a means of **concretely communicating feature enhancement requirements** to SCA tool vendors, and as **a criterion for tool selection by each development team**.

What is Automotive SBOM?

Automotive SBOM = SBOM best practice in Automotive Industry

- **Data Fields**
Definition of Data Items to be Handled as SBOM
- **Automation Support**
Definition of Document Format, Format of Each Data, and Implementation Examples
- **Practice and Process**
Definition of Operational Procedures for Requesting, Generating, and Using SBOM

Data Fields

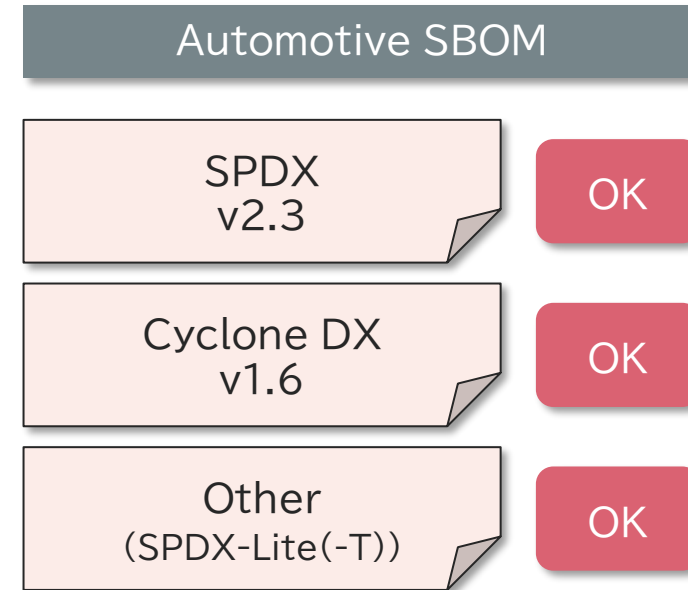
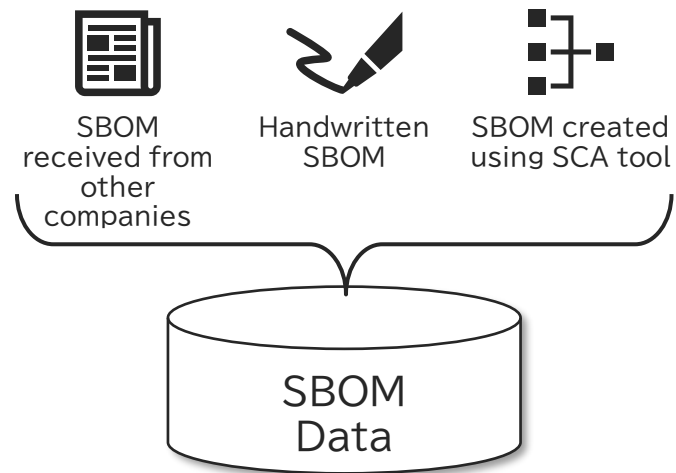
Data Fields

Defined as the Minimum Data Set Required to Understand the Configuration Information of the Target Software

#	Name of Data Field	Required	Description	Example
1	SBOM Metadata			
1-1	SBOM Author Name	☐	Name of the entity that created the SBOM	"TOYOTA MOTOR CORPORATION"
1-2	SBOM Timestamp	☐	Date and time when the SBOM was created or updated	"2025-01-24T22:31:37Z"
1-3	SBOM Type	☐	Information to identify when and what types of software the SBOM was generated from	(Using SBOM type like "Build" etc.. which is defined by CISA at USA)
1-4	SBOM Primary Component	☐	Information to identify the components that are primarily represented in the SBOM	(Express by using Component entities#2)
2	Component Attribute			
2-1	Component Name	☐	Name of the component	"Apache Tomcat"
2-2	Component Version	☐	Version information of the component	"10.0.5"
2-3	Component Supplier Name	☐	Name of the entity that provided the component	Woven by Toyota, Inc.
2-4	Component Relationship	☐	Information to identify dependencies between components	(Express following document format)
2-5	Component Unique Identifier	☐	Information to uniquely identify the component	"pkg:github/apache/tomcat"
2-6	Component File Name	-	Name of the file of the component	"tomcat-10.0.5.tar.gz"
2-7	Component Download Location	-	URL from which the component is downloaded	"https://github.com/apache/tomcat/...../10.0.5.tar.gz"
2-8	Component Declared License	-	License declared by the component's author	Apache-2.0"
2-9	Component Concluded License	☐	License that the SBOM author concluded applies to the component	"Apache-2.0"
2-10	Component Cryptographic Hash	-	Hash value of the component	"SHA256: 525.....5a8"
2-11	Component Copyright Notice	☐ *	Copyright notice attached to the component	"Copyright © 1999-2022, The Apache Software Foundation"
2-12	Component External Document References	☐	Information for external references to other SBOM documents	(Express following document format)

Recommended Data Format

Automation
Support

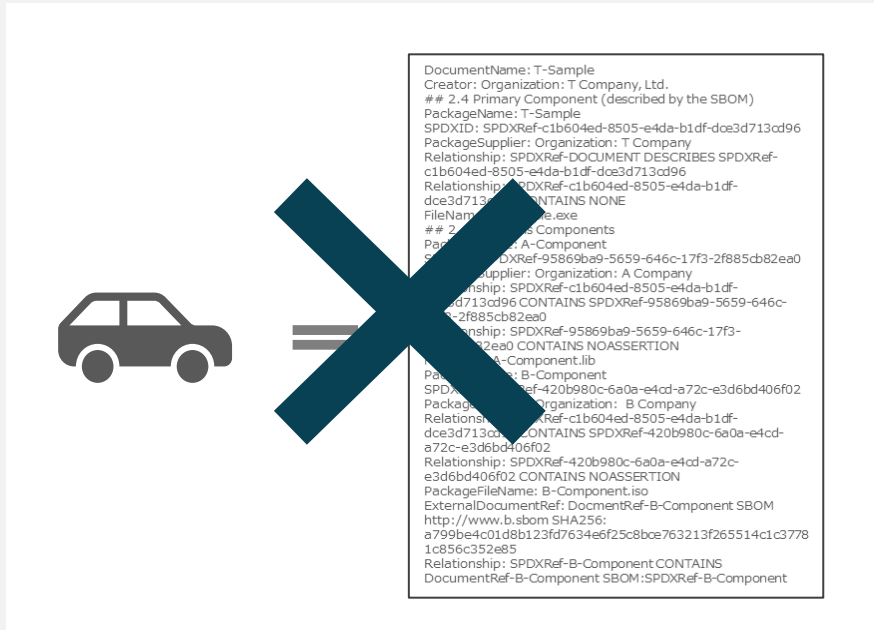


Any format is acceptable as long as it adheres to the Automotive SBOM data fields and practices and processes

Practice and Process

Practice and Process

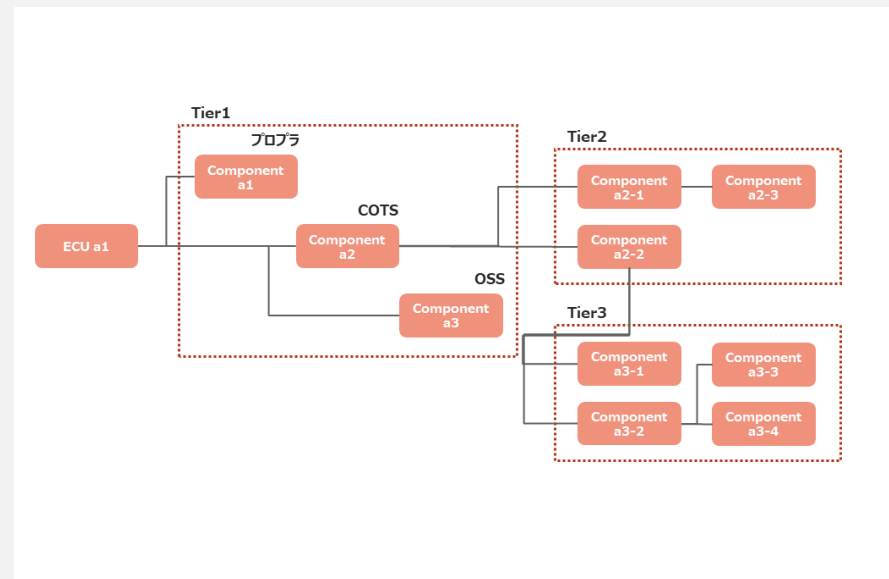
In a Single SBOM File



Disadvantages

- SBOM file becomes too large
- Time-consuming to recreate the entire SBOM file every time you update your software

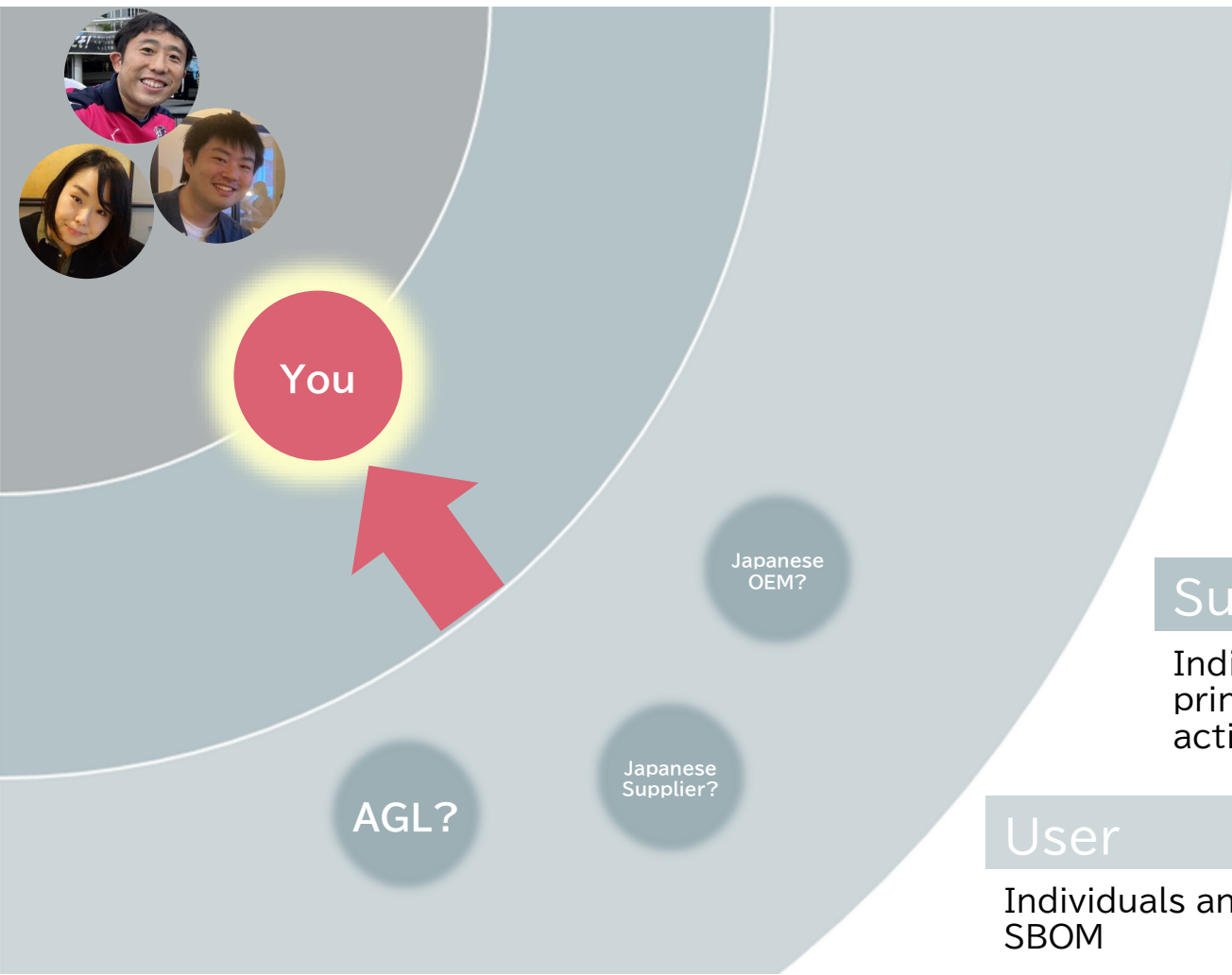
Binding SBOMs Using External Reference Expressions



Advantages

- Easy to handle as a collection of small SBOMs
- High maintainability as there is no need to recreate the entire system every time a software update is made

Join us



Multiple Approaches to Support...

Contributor

Individuals and organizations actively engaged in the evaluation, advocacy, and utilization of Automotive SBOM

Supporter

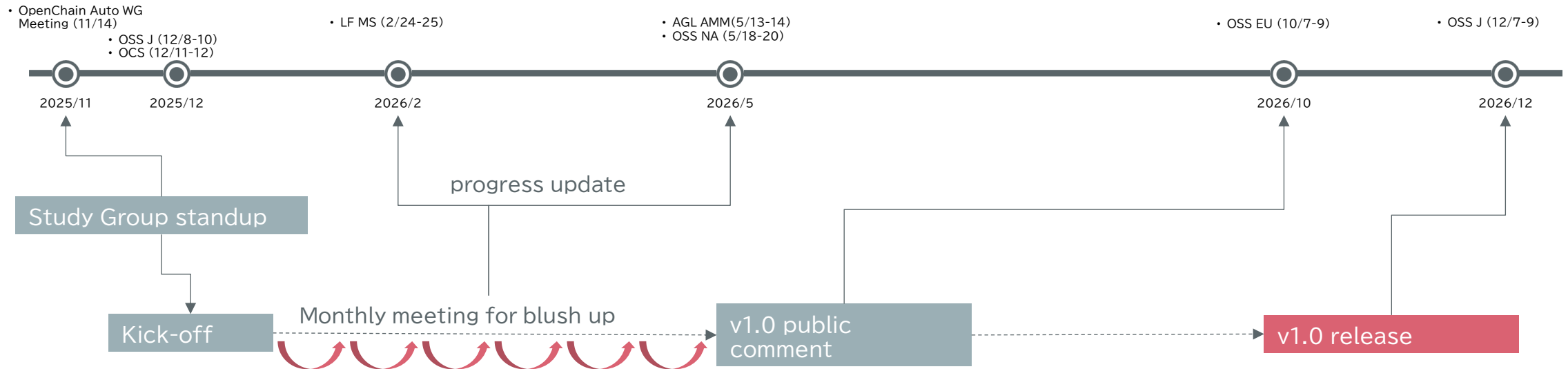
Individuals and organizations that understand the principles of Automotive SBOM and support contributor activities

User

Individuals and organizations that adopt and leverage Automotive SBOM

Roadmap

Events



Your Cooperation, Please

- Review our draft and give us any feedback
- Join our kick-off meeting (time and location will be announced)

Thank you

Please provide feedback/comments to Ayumi.

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